# LNA Notes

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## Desain Drawing

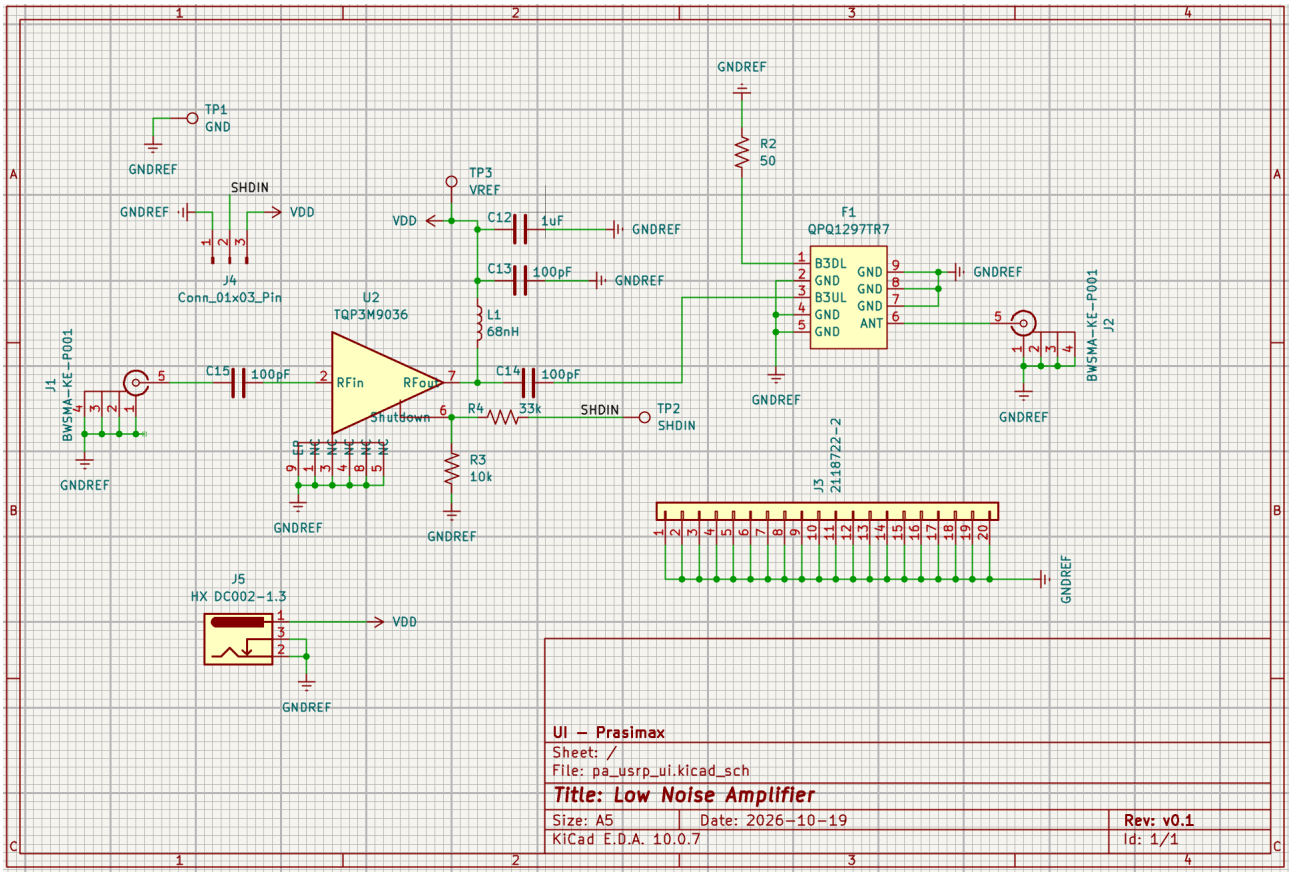



*Figure 1: Rangkaian LNA*

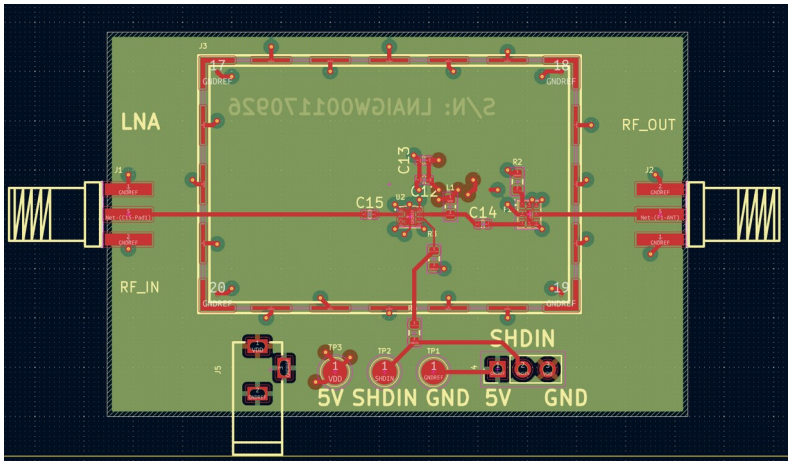



*Figure 2: Rangkaian PCB.*

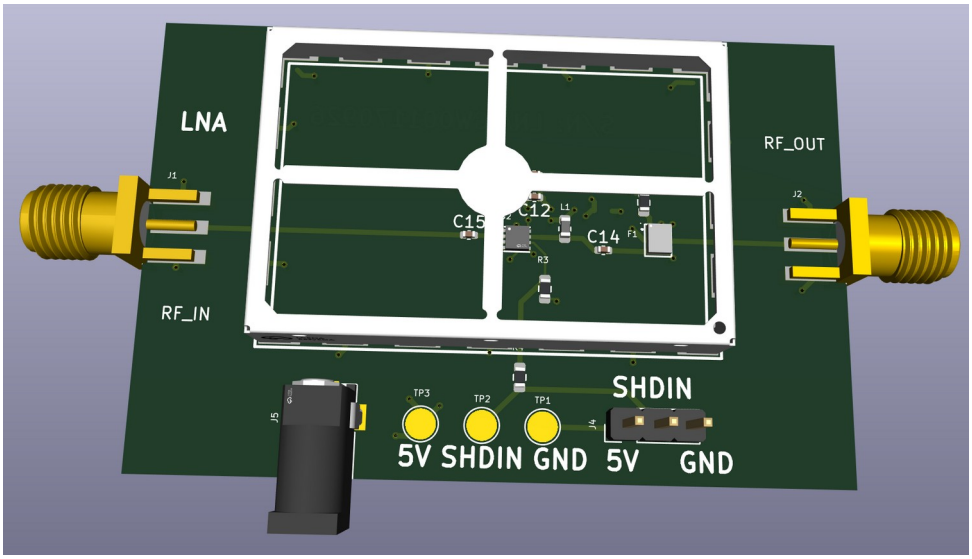



*Figure 3: Gambar 3D dari LNA*

Figure 1 adalah rangkaian Low Noise Amplifier, Figure 2 adalah bentuk PCB untuk Low Noise Amplifier. Perhatikan bahwa cara untuk mencatu amplifier tersbuet terbuka 3 pilihan yaitu dengan barrel jack seperti J5, pad untuk solder seperti test point dan pin header tulang ikan seperti pada konektor j4.

| Reference | Qty | Value | price | lcsc_id | Total |
|---|---|---|---|---|---|
| C12 | | 11uF | | 0.01C52923 | 0.01 |
| C13,C14,C15 | | 3100pF | | 0C1546 | 0.01 |
| F1 | | 1QPQ1297TR7 | | 8.01C471170 | 8.01 |
| J1,J2 | | 2BWSMA-KE-P001 | | 0.41C496550 | 0.82 |
| J3 | | 12118722-2 | | 2.01C5371638 | 2.01 |
| J4 | | 1Conn_01x03_Pin | | | 0 |
| J5 | | 1HX DC002-1.3 | | 0.08C5184827 | 0.08 |
| L1 | | 168nH | | 0.04C6364697 | 0.04 |
| R2 | | 1 | 50 | 0.03C849792 | 0.03 |
| R3 | | 110k | | 0.01C25804 | 0.01 |
| R4 | | 133k | | 0C4216 | 0 |
| TP1 | | 1GND | | | 0 |
| TP2 | | 1SHDIN | | | 0 |
| TP3 | | 1VREF | | | 0 |
| U2 | | 1TQP3M9036 | | 3.13C920261 | 3.13 |
| | | | | total | 14.14 |

# Performa Claim

Dari Figure satu kita melihat bahwa rangkaian input_rf akan melalui komponent berupa SMA connektor J1 → c15 → U2 → c13 → f1 → j2. Dengan mengetahui karakteristik dari jalur rf yang digunakan kita bisa menentukan perkiraan seperti gain maupun insertion loss dari rangkaian tersebut.

## J1,J2 : BWSMA-KE-P001

5.2mm 50Ω Board Side Inner hole Offset Pin SMA SMD Coaxial Connectors (RF) ROHS.

# F1 : QPQ1297

The QPQ1297 is a high-performance Bulk Acoustic Wave (BAW) Duplexer designed for Band 3 uplink / downlink applications.

The QPQ1297 provides low insertion loss and high rejection making it an ideal choice for Small Cells. This duplexer is housed in a compact, RoHs compliant 2.00 mm x 2.50 mm x 0.93 mm surface mount pacakge (SMP).

The QPQ1297 is part of Qorvo's extensive portfolio of RF BAW and SAW filters.

3: FD-LTE, 5MHz 16QAM, PAR =8dB

## Electrical Specifications [1] – Band 3 Uplink

Test conditions unless otherwise specified. Temperature Range: −40 to +95 °C

| Parameter [2] | Conditions | Min | Typ. [3] | Max | Units |
|---|---|---|---|---|---|
| Passband | | 1710 | – | 1785 | MHz |
| Max Insertion Loss | 1710 – 1785 MHz | – | 2.3– | 4.1 | dB |
| Average Insertion Loss | 1710 – 1715 MHz | – | 1.8 | 2.6 | |
| | 1715 – 1780 MHz (Over any 5 MHz bandwidth) | – | 1.6 | 2.4 | |
| | 1780 – 1785 MHz | – | 2.1 | 3.2 | dB |
| Amplitude Ripple [4] | 1710 – 1785 MHz | – | 0.7 | 2.2 | dB |
| Group Delay Variation | 1710 – 1785 MHz | – | 26.9 | 48.0 | ns |
| | 1710 – 1785 MHz (Over any 5 MHz bandwidth) | – | 11.9 | 25.0 | ns |
| Input / Output VSWR | 1710 – 1785 MHz | – | 1.7:1 | 2.0:1 | |
| Attenuation [5] | 10 – 700 MHz | 30 | 38 | – | dB |
| | 700 – 1000 MHz | 38 | 39 | – | |
| | 1000 – 1400 MHz | 20 | 40 | – | |
| | 1400 – 1600 MHz | 35 | 40 | – | |
| | 1600 – 1685 MHz (Over any 5 MHz bandwidth) | 19 | 28 | – | |
| | 1685 – 1690 MHz (Over any 5 MHz bandwidth) | 10 | 24 | – | |
| | 1805 – 1850 MHz (Over any 5 MHz bandwidth) | 44 | 56 | – | |
| | 1850 – 1855 MHz (Over 5 MHz bandwidth) | 48 | 53 | – | |
| | 1855 – 1875 MHz (Over any 5 MHz bandwidth) | 47 | 50 | – | |
| | 1875 – 1880 MHz (Over 5 MHz bandwidth) | 47 | 50 | – | |
| | 1880 –1920 MHz | 38 | 48 | – | |
| | 1920 –1980 MHz | 30 | 45 | – | |
| | 1980 – 2110 MHz | 38 | 46 | – | |
| | 2110 – 2170 MHz | 40 | 50 | – | |
| | 2170 – 2690 MHz | 32 | 39 | – | |
| | 2690 – 3400 MHz | 13 | 19 | – | |
| | 3400 – 3800 MHz | 6 | 9 | – | |
| | 3800 – 5150 MHz | 2 | 4 | – | |
| | 5150 – 6000 MHz | 7 | 9 | – | |
| EVM | 1712.5 – 1782.5 MHz | – | 3 | | % |
| Source/Load Impedance [6] | Single-ended | – | 50 | – | Ω |

*Figure 4: tabel untuk suhu -40 hingga +95 Celcius*

Detail bisa dicheck https://www.qorvo.com/products/r/ra005518

# LNA IC



## Typical Performance – TQP3M9036-PCB VDD = 5 V, 25ºC

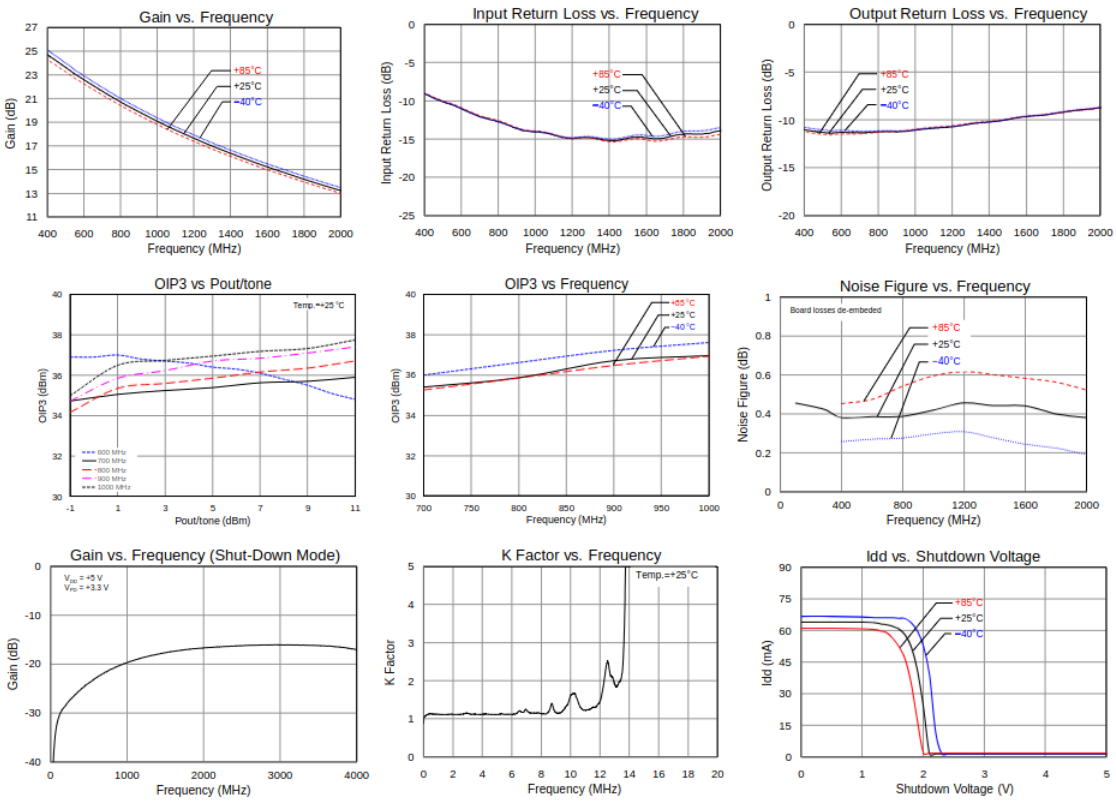
| Parameter | Conditions | Typical Value | | | | | Units |
|---|---|---|---|---|---|---|---|
| Frequency | | 600 | 700 | 900 | 1500 | 2000 | MHz |
| Gain | | 22.5 | 20.8 | 19.1 | 15.1 | 13 | dB |
| Input Return Loss | | -11 | -11.2 | -12.7 | -14.3 | -13.5 | dB |
| Output Return Loss | | -11.5 | -10.4 | -11.0 | -11.1 | -8.6 | dB |
| Output P1dB | | +23.5 | +23.1 | +23.2 | +23.2 | +23.2 | dBm |
| Output IP3 | Pout= +5 dBm/tone, Δf=1 MHz | +36 | +34.2 | +35.2 | +36 | +37 | dBm |
| Noise figure (1) | | 0.38 | 0.38 | 0.4 | 0.44 | 0.52 | dB |

Notes:

1. Noise figure data shown in the table above is de-embedded from the eval board loss.

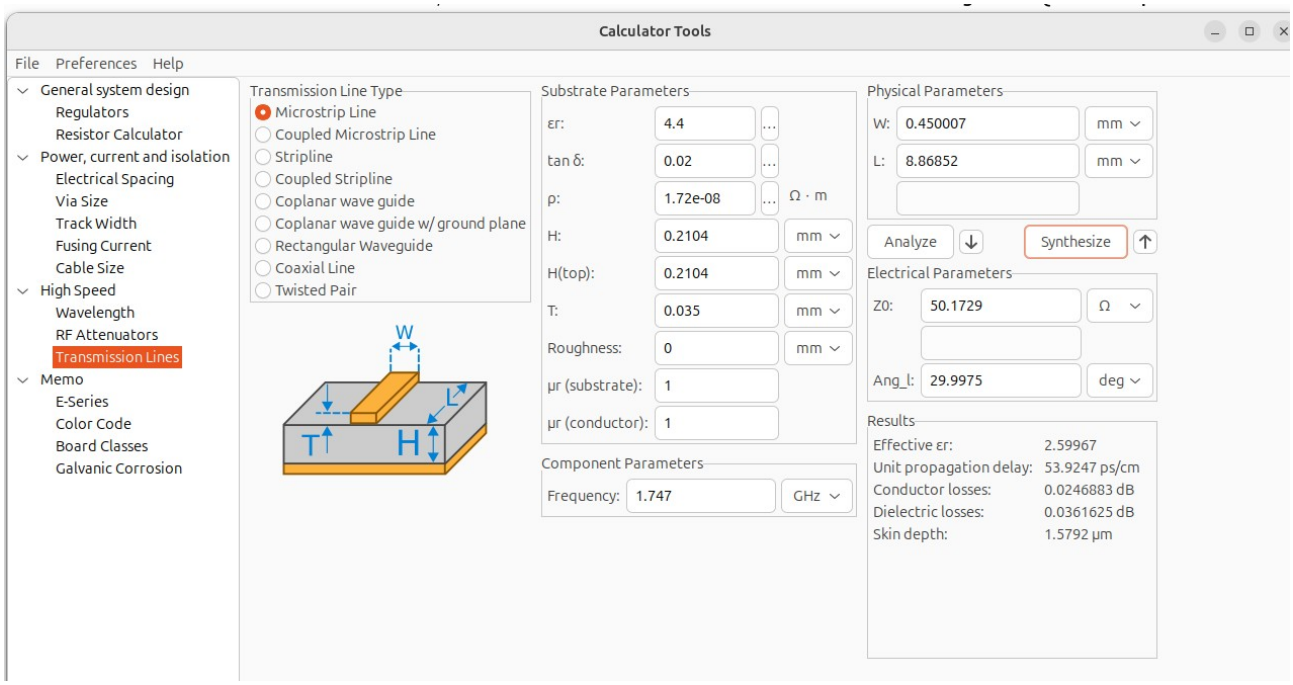
## Performance Plots – TQP3M9036-PCB VDD = 5 V

Test conditions unless otherwise noted: $V_{DD}$ = +5 V, $I_{DD}$ = 68 mA, $T_{CASE}$ = +25°C, 50 Ω system

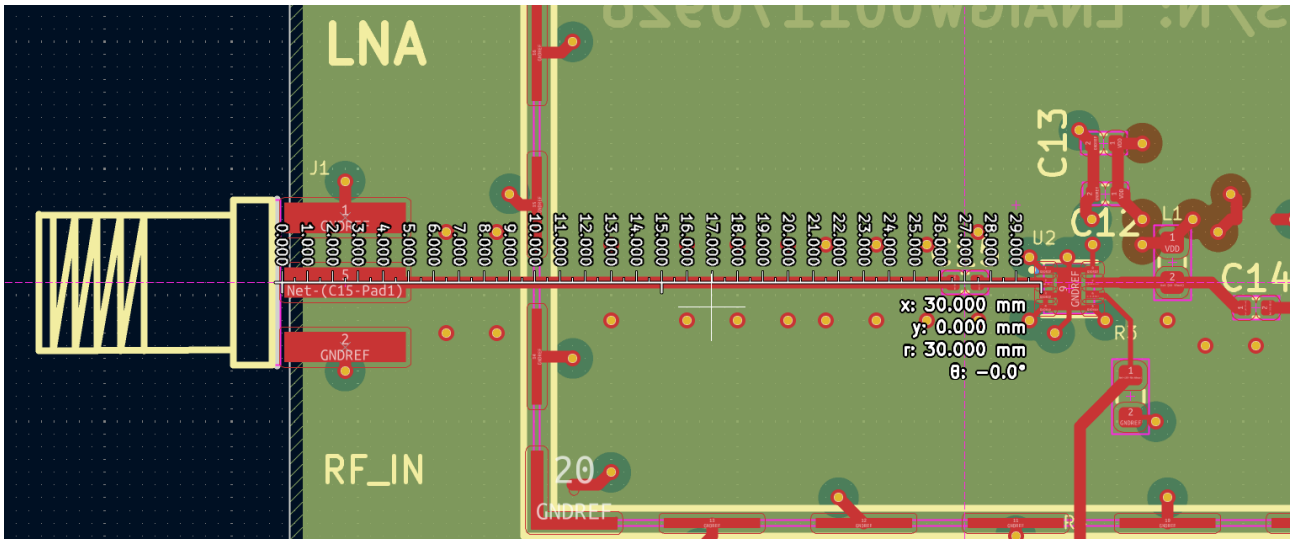



Detail bisa check di https://www.qorvo.com/products/r/ra004882

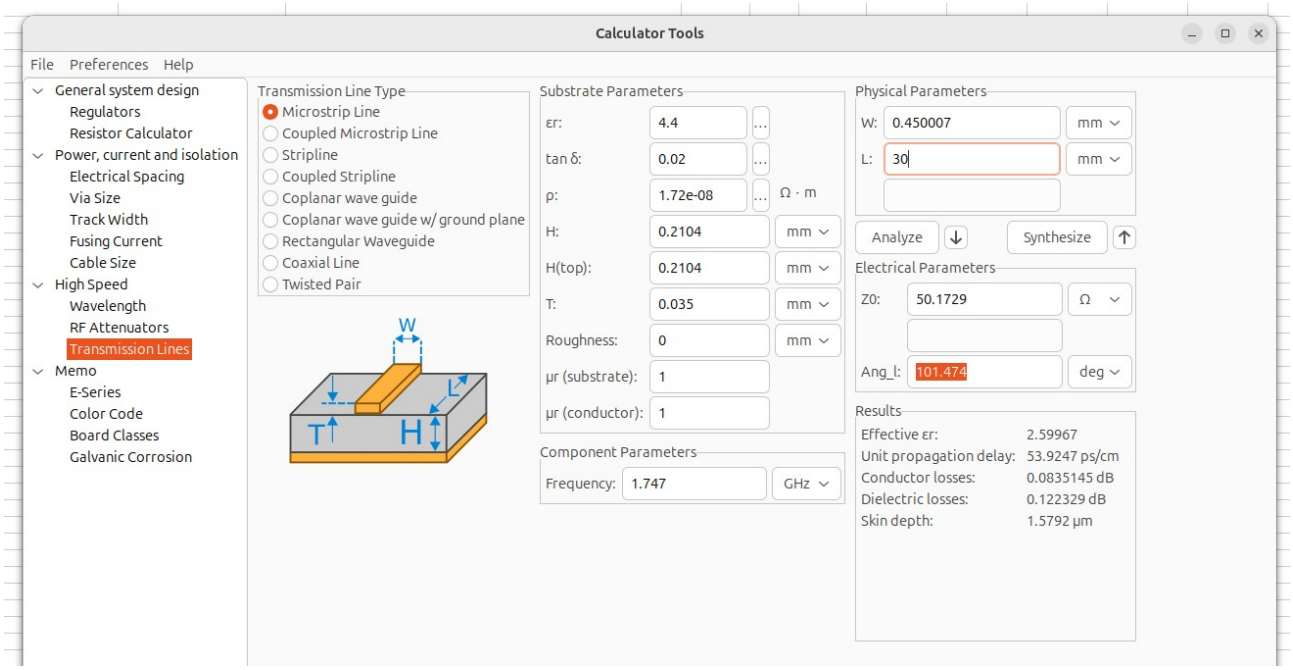
# PCB LNA



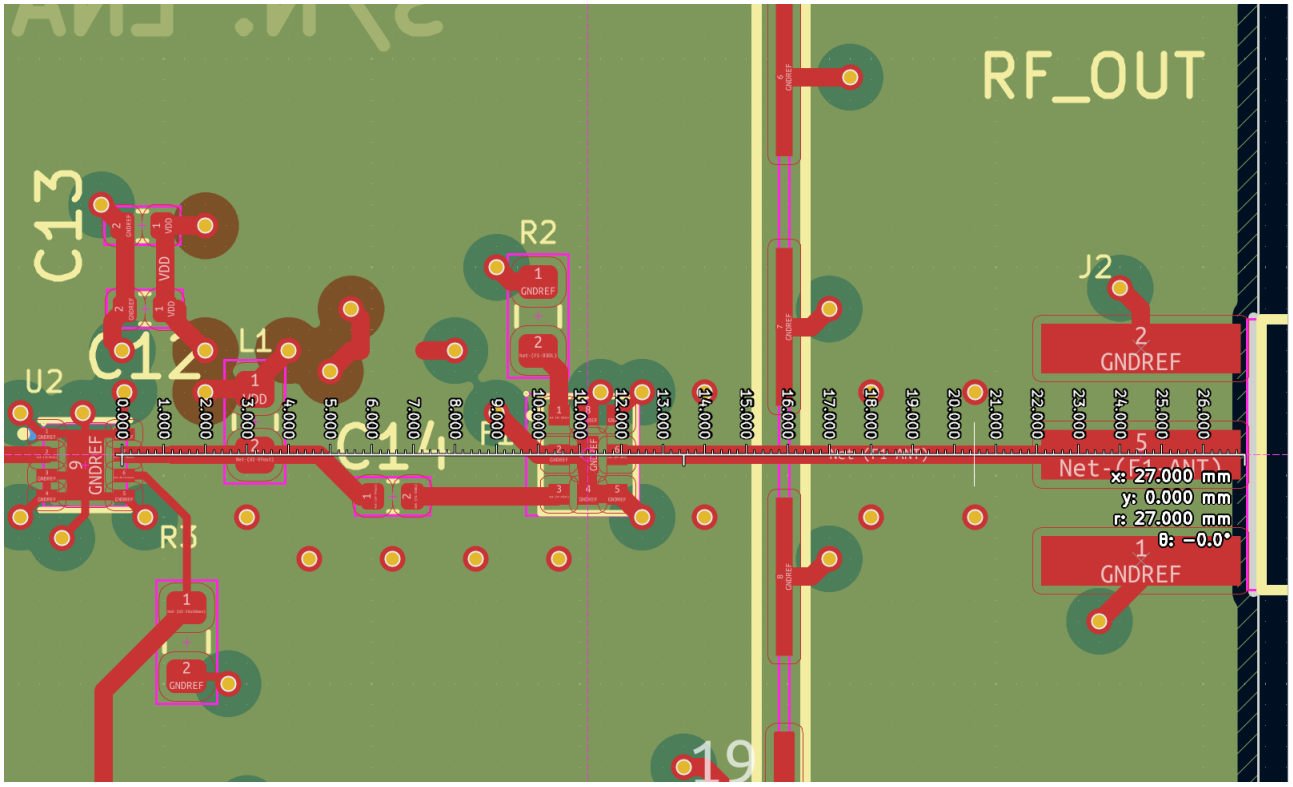

Z0 pada frequensi 1.747 GHz adalah 50.1729 Ohm



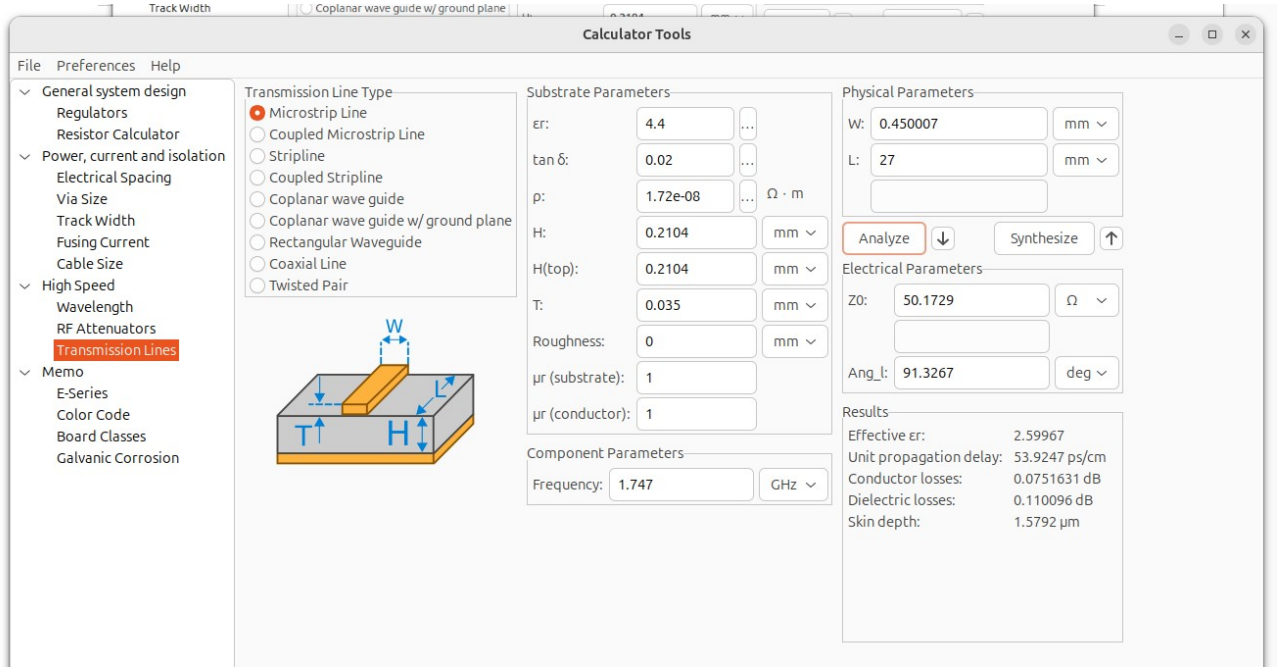

30 mm



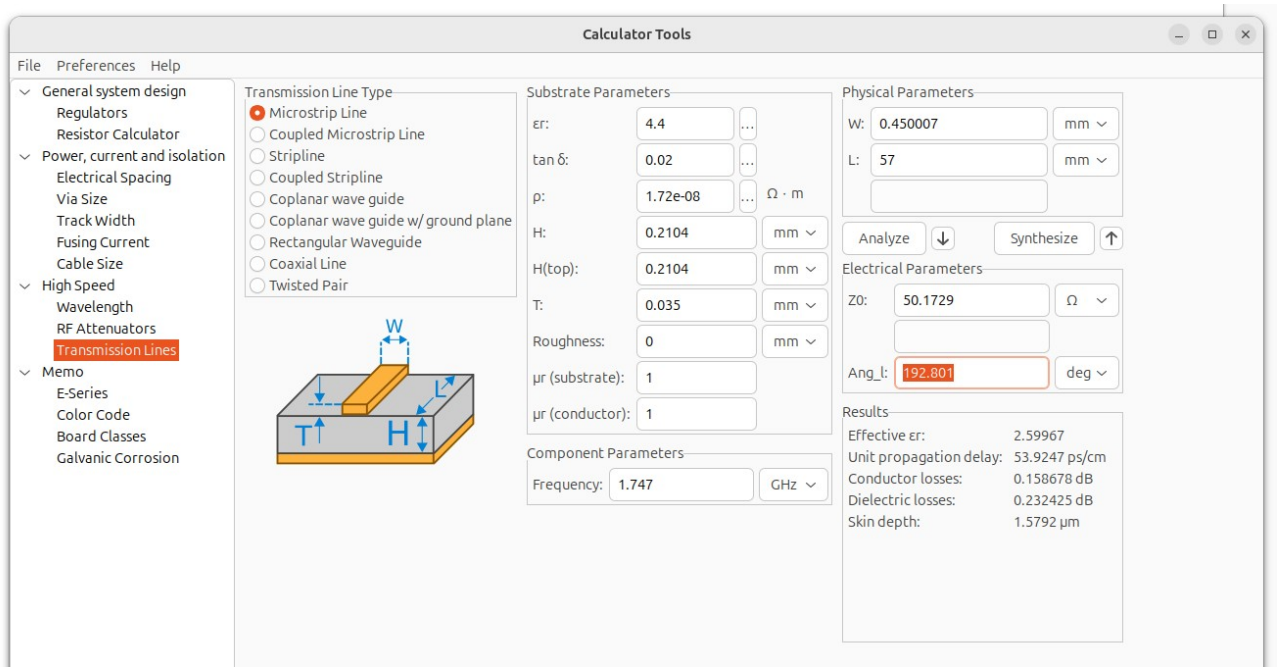

phase delay antara rf input dan input LNA 101.47 degree




27 mm antara pa dan rf out.




Phase delay antara lna dan rf output 91.3267 degree




total delay phasa tanpa memperhitungan delay LNA dan delay filter dalah 192 degree

$$Z_0 = 50.1729\,\text{ohm}$$

$$Z_L = 50\,\text{ohm}$$

$$Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$Gamma = \frac{50 - 50.1729}{50 + 50.1729}$$

$$Gamma = \frac{-0.1729}{100.1729}$$

$$|(Gamma)| \approx 0.0017257$$

$$VSWR = \frac{1 + |(Gamma)|}{1 - |(Gamma)|}$$

$$VSWR = \frac{1 + 0.0017257}{1 - 0.0017257}$$

$$VSWR \approx 1.00346$$

$$RL = -20\log\left(|(Gamma)|\right)$$

$$RL = -20\log\left(0.0017257\right)$$

$$RL \approx 55.26\,\text{dB}$$